Housing, Fertility, and Demographic Renewal

A two-generation model

Environment

Agents. Time is discrete and one period is one generation. At date t, the economy contains Y_t young households and O_t old households. A young household has type $i = (y_i, b_i)$, drawn from a fixed distribution F of income and liquid wealth. A young renter becomes an old renter, a young owner becomes an old incumbent, and every old household exits at the end of the period. Entry wealth is exogenous: estates are not assigned to particular descendants.

Preferences. A young household values consumption, space net of its children's needs, and completed fertility:

$$u_t^Y(c, h, n) = \log(c - \chi n) + \alpha \log(h - \kappa n) + \vartheta_t \log n. \tag{1}$$

Each child requires $\chi > 0$ units of goods and $\kappa > 0$ units of space. An old household values consumption, housing, and a warm-glow estate:

$$u^O(c^O, h^O, e) = \log c^O + \gamma \log h^O + \omega_B \log e. \tag{2}$$

The estate affects the old household's utility but does not determine the number or wealth of next period's young households.

Housing. There is a fixed stock $\bar{H}$ of divisible floorspace. A renter can occupy at most $\bar{h}^R$, while an owner can occupy at most $\bar{h}^O > \bar{h}^R$. The rental price is r_t and the asset price is P_t. With one-period bond price q, depreciation δ, and property-tax rate τ_t^p, no arbitrage requires:

$$r_t = \left(q^{-1} + \delta + \tau_t^p\right) P_t - P_{t+1}. \tag{3}$$

The familiar stationary relation $r = (q^{-1} - 1 + \delta + \tau^p)P$ is the special case $P_{t+1} = P_t$.

Housing institutions. A buyer can finance at most a share $\phi \in (0, 1)$ of the purchase price and must therefore post $(1 - \phi)P_t h$ from current liquid wealth. Selling in old age realizes a nominal capital gain, while death receives stepped-up basis. If nominal prices grow at rate ϖ, the real tax due per unit sold at date t is:

$$\ell_t = \tau^g \left[P_t - \frac{P_{t-1}}{1 + \varpi} \right]_+ . \tag{4}$$

At a stationary real price this becomes $\bar{\ell} = \tau^g P \varpi/(1 + \varpi)$, exactly the wedge in the July model. Tax receipts accrue to the government; any rebate is absorbed into the exogenous nonhousing resources y_i and does not alter the marginal conditions below.

Demography. One child produces $\nu > 0$ future young households after survival and household formation. The scalar ν is the simple model's counterpart of the quantitative model's person-to-household conversion.

Household problems

Young households

Conditional on tenure $m \in \{R, O\}$, a young household chooses consumption, housing, completed fertility, and bond expenditure:

$$\begin{aligned} W_t^m(i) = \max_{c,h,n,b'} \; & u_t^Y(c, h, n) + \beta \left[\mathbf{1}\{m = R\} V_{t+1}^R(b_{t+1}) + \mathbf{1}\{m = O\} V_{t+1}^O(b_{t+1}, h) \right] \\ \text{subject to} \quad & c + r_t h + b' = y_i + b_i, \\ & h \le \bar{h}^m, \qquad b' \ge 0, \\ & \mathbf{1}\{m = O\}(1 - \phi)P_t h \le b_i, \\ & b_{t+1} = b'/q - \mathbf{1}\{m = O\} r_{t+1} h. \end{aligned} \tag{5}$$

The household chooses the tenure with the higher value. Its effective family-housing caps are:

$$H_t^R = \bar{h}^R, \qquad H_t^O(i) = \min\left\{ \bar{h}^O, \frac{b_i}{(1 - \phi)P_t} \right\}. \tag{6}$$

Thus current liquid wealth, rather than the ability to service the flow cost, can prevent a household from occupying family-sized owner housing.

Old households

An old incumbent arrives with liquid wealth b, a house H^0, and the tax basis established when young. She chooses how much of the house to retain:

$$V_t^O(b, H^0) = \max_{c^O, h^O, e} u^O(c^O, h^O, e) \tag{7}$$
$$\text{subject to} \quad c^O + e + r_t h^O = b + r_t H^0 - \ell_t(H^0 - h^O), \qquad 0 \le h^O \le H^0.$$

At an interior retention choice, her marginal value of retained space is $r_t - \ell_t$. An old renter instead solves the same utility problem subject to $c^O + e + r_t h^O = b$ and values marginal housing at r_t.

The old owner's budget can be written as $c^O + e + (r_t - \ell_t)h^O = b + (r_t - \ell_t)H^0$. Define her effective resources by $\Omega_t^O = b + (r_t - \ell_t)H^0$ and let $A = 1 + \gamma + \omega_B$. When the retention limit is slack and $r_t > \ell_t$, the log problem has the closed-form allocation:

$$c_t^O = \frac{\Omega_t^O}{A}, \qquad h_t^O = \frac{\gamma \Omega_t^O}{A(r_t - \ell_t)}, \qquad e_t = \frac{\omega_B \Omega_t^O}{A}. \tag{8}$$

The old incumbent behaves as if retained housing carried price $r_t - \ell_t$. For an old renter, equation (8) applies with $\Omega_t^R = b$, housing price r_t, and no carried house.

Young first-order conditions

The link between the two periods is clearest after substituting the old problem into the young problem. A young owner's next-period effective resources are:

$$\Omega_{t+1}^O = \underbrace{\left(\frac{b'}{q} - r_{t+1} h\right)}_{b_{t+1}} + \underbrace{(r_{t+1} - \ell_{t+1})h}_{\text{value of carried housing}} = \frac{b'}{q} - \ell_{t+1} h. \tag{9}$$

The future housing service cancels the flow-equivalent liability carried into old age; only the tax on a sale remains. A young renter instead has $\Omega_{t+1}^R = b'/q$.

Let $\Lambda = \beta A$ be old-age claims on one unit of young marginal utility. At an interior saving choice, the Euler equation and equation (9) imply:

$$b' = \Lambda(c - \chi n) + \mathbf{1}\{m = O\} q \ell_{t+1} h. \tag{10}$$

The buyer saves the ordinary old-age claim plus the discounted future tax bill attached to the house. At $b' = 0$, the corresponding complementary-slackness condition replaces equation (10).

Define residual consumption $x = c - \chi n$, residual adult space $s = h - \kappa n$, and the tenure-specific effective housing price $r_t^R = r_t$ and $r_t^O = r_t + q\ell_{t+1}$. Substituting the saving rule into the young budget gives the lifetime resource identity:

$$y_i + b_i = (1 + \Lambda)x + r_t^m s + \big(\chi + \kappa r_t^m\big)n. \tag{11}$$

Thus $\chi + \kappa r_t^m$ is the unconstrained full price of a child. When the housing and financing constraints are slack, the log problem allocates lifetime resources according to:

$$\begin{aligned} D_t &= 1 + \alpha + \vartheta_t + \Lambda, \quad x_t^m = \frac{y_i + b_i}{D_t}, \\ n_t^m &= \frac{\vartheta_t(y_i + b_i)}{D_t(\chi + \kappa r_t^m)}, \qquad h_t^m = \frac{\alpha(y_i + b_i)}{D_t r_t^m} + \kappa n_t^m. \end{aligned} \tag{12}$$

Higher housing costs reduce fertility because each child requires κ units of space; they also reduce adult housing directly.

To characterize constrained households, let λ_t^m be the multiplier on the young budget, η_t^m the multiplier on the tenure-specific size limit, and μ_t the multiplier on the owner's down-payment constraint. The goods value of relaxing the active housing constraints is:

$$\zeta_t^R = \frac{\eta_t^R}{\lambda_t^R}, \qquad \zeta_t^O = \underbrace{\frac{\mu_t}{\lambda_t^O}(1-\phi)P_t}_{\text{financing gap}} + \underbrace{\frac{\eta_t^O}{\lambda_t^O}}_{\text{owner-size gap}}. \tag{13}$$

The housing first-order condition then identifies the total access gap from the household's bundle:

$$\frac{\alpha(c_t^m - \chi n_t^m)}{h_t^m - \kappa n_t^m} = r_t^m + \zeta_t^m, \qquad \zeta_t^m \geq 0. \tag{14}$$

An unconstrained household has $\zeta_t^m = 0$; a constrained household values one more unit of space above its effective price by exactly ζ_t^m.

Finally, the fertility first-order condition prices the marginal child in units of current goods:

$$\frac{\vartheta_t(c_t^m - \chi n_t^m)}{n_t^m} = \chi + \kappa\big(r_t^m + \zeta_t^m\big). \tag{15}$$

A binding housing constraint therefore acts as a household-specific tax of $\kappa\zeta_t^m$ on fertility.

Demography, markets, and transition equilibrium

Let $\bar{n}_t$ and $\bar{h}_t^Y$ denote average fertility and housing among the young, integrating the policies in equation (5) over F. Let $\bar{h}_t^O$ denote average housing among the old distribution generated by the preceding young cohort. Births are therefore:

$$B_t = Y_t \bar{n}_t. \tag{16}$$

The baseline closed economy and its open-market extension are nested in the cohort law:

$$Y_{t+1} = M_{t+1} + \rho\nu B_t, \qquad O_{t+1} = Y_t. \tag{17}$$

The closed national economy sets $M_t = 0$ and $\rho = 1$. In the open extension, M_t is an outside inflow and $\rho \in [0, 1]$ is the retention rate of locally born future households.

Housing clears at every date rather than only at an endpoint:

$$Y_t \bar{h}_t^Y + O_t \bar{h}_t^O = \bar{H}. \tag{18}$$

The clearing condition determines the price path jointly with household expectations and the inherited cohort distribution.

Definition 1 (Transition equilibrium)**.** Given the initial cohort masses, old-household distribution and tax bases, a path of primitives, and a terminal condition when one is available, a transition equilibrium is a sequence of prices, household policies, distributions, births, and cohort masses such that: (i) young and old choices solve their problems given the anticipated price path; (ii) the old distribution is the push-forward of the preceding young cohort's tenure, housing, and saving choices; (iii) equation (17) generates the next young cohort; (iv) housing clears according to equation (18) at every date; and (v) asset pricing and government accounting hold at every date.

A transition equilibrium is a dated sequence, not by definition movement between two steady states. Convergence to a new steady state is an additional boundary condition that can be imposed only when that steady state exists.

The two-cohort law also makes demographic momentum explicit. In the closed economy, define the effective reproduction rate $g_t = \nu\bar{n}_t$ and the inherited old-young ratio $\mu_t = O_t/Y_t$. Total adult-household population $S_t = Y_t + O_t$ then obeys:

$$S_{t+1} = Y_t(1 + g_t), \qquad S_t = Y_t(1 + \mu_t), \qquad S_{t+1} > S_t \iff g_t > \mu_t. \tag{19}$$

Hence below-replacement fertility can coexist with population growth when $\mu_t < g_t < 1$. Starting from an exact steady state, however, $\mu_t = 1$, so an immediate fall below replacement reduces total population in the next period.

Steady states

Under constant primitives, let $\mathcal{N}(P)$ denote average fertility per young household in the stationary household problem, and let $\mathcal{H}^Y(P)$ and $\mathcal{H}^O(P)$ denote stationary average housing by the two cohorts.

Proposition 1 (Closed steady state)**.** *Suppose the stationary household problem is well defined. A positive closed steady state exists only at a price $P^* > 0$ satisfying:*

$$\nu\mathcal{N}(P^*) = 1. \tag{20}$$

For every such price, the associated cohort masses are:

$$Y^* = O^* = \frac{\bar{H}}{\mathcal{H}^Y(P^*) + \mathcal{H}^O(P^*)}. \tag{21}$$

If $\mathcal{N}(P)$ is continuous and crosses $1/\nu$ exactly once, the positive closed steady state is unique.

Proof. Stationarity of the aging law gives $O^* = Y^*$. With $M = 0$ and $\rho = 1$, the young-cohort law becomes $Y^* = \nu Y^*\mathcal{N}(P^*)$. Dividing by the positive mass Y^* gives equation (20). Substituting $O^* = Y^*$ into housing clearing gives equation (21). A unique crossing of the replacement schedule gives a unique price, after which housing clearing gives a unique positive scale. □

Demographic replacement selects the stationary price; housing clearing then selects how many households can live at that price. If $\nu\mathcal{N}(P) < 1$ at every positive price, the model has no positive closed steady state. A finite transition can still be computed, but it should not be described as movement to a second positive steady state.

With constant outside inflow $M > 0$, the open steady-state cohort mass instead satisfies:

$$Y^* = O^* = \frac{M}{1 - \rho\nu\mathcal{N}(P^*)}, \qquad \rho\nu\mathcal{N}(P^*) < 1. \tag{22}$$

Together, equations (18) and (22) determine the open-market price and scale. Thus an open local economy can remain stationary with below-replacement fertility because outside inflows fill the replacement gap.

Suppose the economy begins at an old steady state and ϑ_t then falls permanently. On impact, Y_t and O_t are inherited, while prices, tenure, housing, and fertility respond. Fertility changes Y_{t+1} through equation (17); the new cohort changes subsequent housing demand and prices. If a new root of equation (20) exists, convergence to it defines a transition between steady states. If it does not exist, the equilibrium remains a nonstationary demographic path.

One implication is immediate. Average fertility in every positive closed steady state equals $1/\nu$. Housing policy can change fertility during the transition, the distribution of fertility across households, and the stationary population scale, but it cannot permanently change average fertility across closed steady states without changing survival or household formation summarized by ν.

Intergenerational housing misallocation

The July reallocation result can be stated at any date of the transition. Hold the price path, cohort masses, tenure assignments, and all other household allocations fixed. Consider a young owner whose financing constraint binds and whose owner-size cap is slack, and an old incumbent at an interior retention choice. Moving one unit of housing from the incumbent to the young buyer generates goods-equivalent surplus:

$$\underbrace{\left(r_t + q\ell_{t+1} + \zeta_t^{O,F}\right)}_{\text{young buyer's marginal value}} - \underbrace{(r_t - \ell_t)}_{\text{old incumbent's marginal value}} = \zeta_t^{O,F} + q\ell_{t+1} + \ell_t > 0. \tag{23}$$

In a stationary equilibrium, $\ell_t = \ell_{t+1} = \bar{\ell}$, so equation (23) becomes the July surplus formula $\zeta^{O,F} + q\bar{\ell} + \bar{\ell}$.

This is a conditional date-t allocation result, not a dynamic welfare theorem. A policy that reallocates housing also changes fertility, future cohort sizes, and equilibrium prices. Evaluating that policy requires welfare weights over present and future cohorts and an explicit decision about whether the utility of unborn children is counted separately from parental utility.